

SYSTEMATIC REVIEW

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Physical therapy versus conventional treatment for grade I and II acute ankle sprains: trial sequential analysis and meta-analysis

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Abstract

Background This systematic review and meta-analysis evaluated the effectiveness of physiotherapy compared with conventional therapies for grade I and II acute ankle sprains.

Methods Randomized controlled trials were identified through a comprehensive search of PubMed, Embase, and Web of Science databases. Eligibility criteria were defined using the PICOS framework. Analyses included forest plot analysis, subgroup analysis, sensitivity analysis, publication bias analysis, and Trial Sequential Analysis (TSA) using Review Manager (version 5.4), StataMP 17, and TSA 0.9.5.10 beta software.

Results Ten RCTs involving 565 participants met the inclusion criteria. The physical therapy group had significantly lower post-treatment visual analog scale (VAS) scores for pain compared to the conventional treatment group (weighted mean difference [WMD] = -0.46, 95% CI = -0.90 to -0.01, $P=0.04$). On subgroup analysis, VAS scores at rest were also significantly lower in the physical therapy group compared to the conventional treatment group (WMD = -0.34, 95% CI = -0.67 to -0.01, $P=0.04$).

Conclusions Physical therapy may offer superior pain relief compared with conventional treatment for grade I and II acute ankle sprains. Further high-quality, adequately powered RCTs are warranted to obtain more robust evidence.

Level of evidence level 1.

Registration CRD42025640304.

Keywords Acute ankle sprain, Physical therapy, Conventional treatment, Functional recovery, Meta-analysis

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Background

Acute ankle sprains are a common sports injury, particularly in activities requiring rapid directional changes and jumping [1–4]. Epidemiological data indicate a rising incidence of acute ankle sprains, likely due to increased sports participation and public health initiatives promoting physical activity [5]. An estimated 2 million ankle sprains occur each year, with the majority being grade I and II sprains [6]. These injuries not only impede daily life and athletic performance but also pose a risk of long-term consequences, including chronic instability necessitating surgery [7], as well as heightened susceptibility to re-injury [8, 9]. Recent advances in imaging [10] and kinematic analyses [11] have enabled better characterization of these injuries, informing appropriate treatment selection. Improvements in arthroscopic repair techniques have shown promising results in severe ankle sprains and chronic ankle instability [12, 13].

Currently, the treatment for grade I and II ankle sprains typically involves physical therapy and conventional treatment. Physical therapy encompasses various approaches, such as exercise therapy, ultrasound therapy, thermotherapy, hydrotherapy, stretching, and functional therapy, but protocols and frequencies vary [14, 15]. Conventional treatments also differ, including RICE (rest, ice, compression, elevation) treatment, standard care, placebo, and plaster immobilization [16]. A study of management practices for acute ankle sprains in the United Kingdom revealed widespread use of elasticated tubular bandage, below-knee casts, and braces [17]. Functional treatment appears to be superior to immobilization for acute ankle sprains [18]. Neuromuscular control training has been shown to mitigate the risk of recurrent ankle sprains [19]. Despite numerous randomized controlled trials (RCTs) involving current treatment modalities, there is a paucity of high-quality research to determine the most effective treatment methods.

Studies on acute ankle sprain treatment vary widely in terms of intervention methods, patient characteristics, and follow-up times [20], resulting in high heterogeneity that complicates direct comparisons and meta-analyses. Furthermore, many studies have small sample sizes, which can lead to inadequate statistical power and increased bias due to the disproportionate influence of individual studies on overall results [21]. Moreover, previous studies have typically focused on single outcome metrics, such as visual analogue scale (VAS) scores for pain or range of movement (ROM) for joint mobility [22]. This limits a comprehensive assessment of treatment effects. A more comprehensive assessment would combine subjective measures like pain with objective indicators like joint mobility [23].

This study aims to address the existing limitations by systematically evaluating the therapeutic efficacy of

physical therapy versus conventional treatment for grade I and II acute ankle sprains. The study seeks to answer three key questions: (1) Does physical therapy more effectively reduce pain compared to conventional treatment, especially in the early and middle stages after treatment? (2) Does physical therapy significantly improve joint mobility, especially dorsiflexion? (3) Is the combined effect of physical therapy in terms of pain relief and joint mobility superior to that of conventional treatment?

Methods

Protocol and registration

This systematic review and meta-analysis adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) 2020 guidelines [24]. The study protocol was registered with PROSPERO (Reg. No. CRD42025640304; available at <https://www.crd.york.ac.uk/PROSPERO/view/CRD42025640304>).

Eligibility criteria

The inclusion and exclusion criteria were defined according to the PICOS framework:

- Population (P): Patients of any sex, aged ≥ 14 years, diagnosed with grade I or grade II acute ankle sprains, as defined by injury severity classification based on ligament damage (grade I: mild stretching without macroscopic tear; grade II: partial ligament tear without complete rupture) [1]. Trials that included grade III injuries were excluded unless separate data for grade I/II could be extracted.
- Intervention (I): Physical therapy interventions, including but not limited to exercise therapy, stretching, ultrasound therapy, thermotherapy, hydrotherapy, functional training, and multimodal rehabilitation programs.
- Comparator (C): Conventional treatment such as RICE (rest, ice, compression, elevation), standard medical care, placebo/sham interventions, cast or plaster immobilization, or no functional exercises.
- Outcomes (O):
Primary outcome: Post-treatment pain intensity measured by the Visual Analogue Scale (VAS).
Secondary outcomes: VAS score at rest, VAS score during exercise, ankle dorsiflexion range of motion (ROM).
- Study Design (S): Randomized controlled trials (RCTs).

Exclusion criteria included non-RCT designs (e.g., observational studies, case series, case reports), conference abstracts without full-text, non-English language studies,

duplicate publications, and trials with incomplete outcome data.

Search strategy

A comprehensive search strategy from database inception to 11 February 2025 was performed across PubMed, Embase, and Web of Science databases. Additionally, the WHO International Clinical Trials Registry Platform

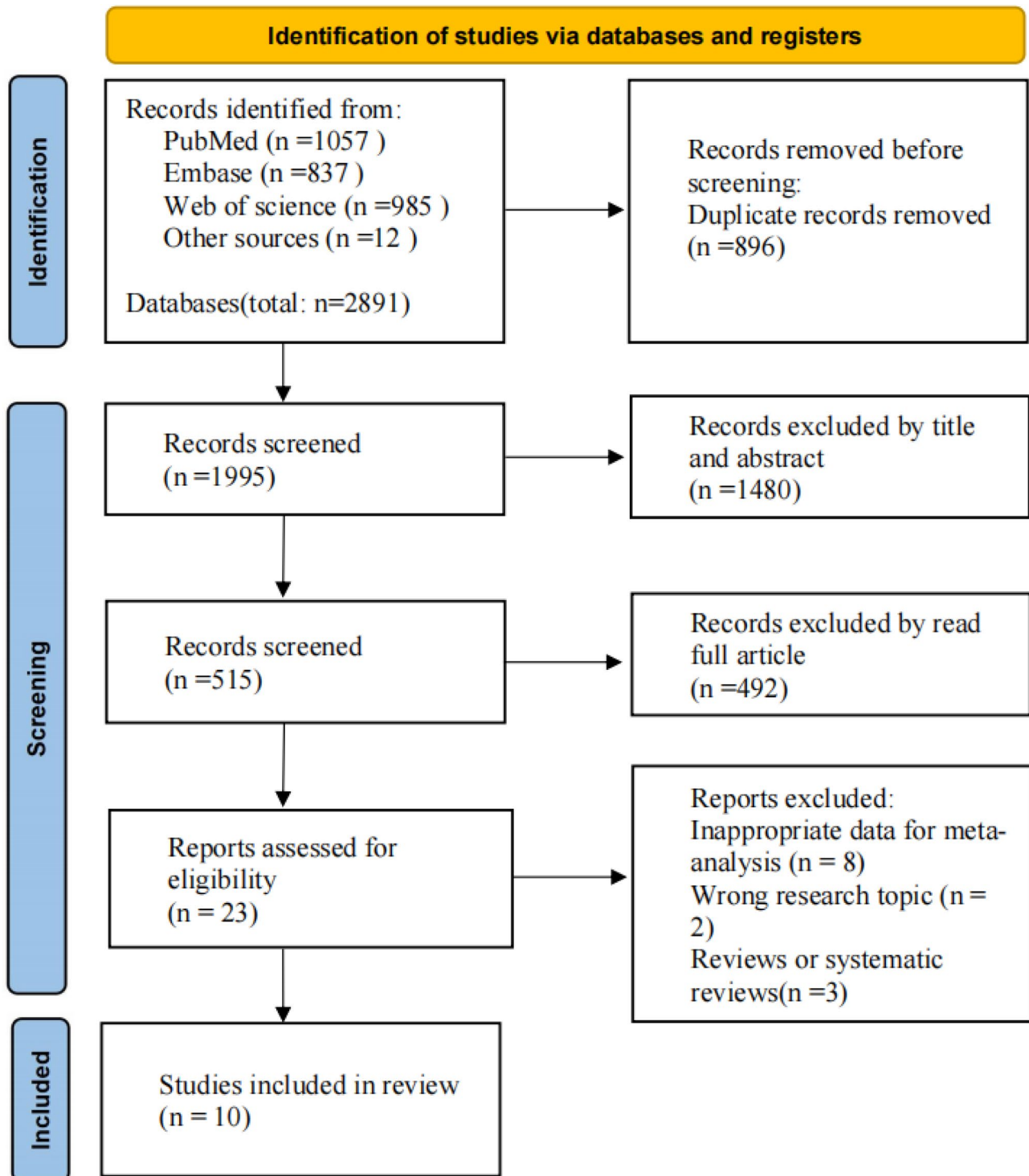


Fig. 1 Schematic illustration of literature search and study-selection criteria

Table 1 Characteristics of patients included in the RCTs

Research	Group	Inclusion Criteria	Intervention	Participants	Age (M/F)	Sex (M/F)	Side (Left/Right)	Type	Follow up
Thomas 2019 [43]	Conventional therapy Physical therapy	1. No mild swelling 2. No fractures 3. No loss of sensation 4. Absence of other injuries 5. BMI not exceeding 40 kg/m ²	Current standard of care Current standard of care plus neuro-muscular electrical stimulation use	11 11	36.0 (8.0) 44.0 (10.0)	4/7 6/5	NA NA	Grade I and II Grade I and II	7 days
Marvam 2024 [15]	Conventional therapy Physical therapy	1. Age between 18 and 30 years 2. BMI between 18 and 25 kg/m ² 3. No fractures	Conventional therapy Aquatic training	15 15	23.80 (3.91) 23.93 (3.26)	6/9 7/8	7/8 9/6	Grade III Grade III	4w-6w
Weerasekara 2016 [31]	Conventional therapy Physical therapy	1. Grade I and II lateral ankle ligament sprains 2. Age between 13 and 50 years	Heat therapy Contrast therapy	46 69	22.2 (6.3) 22.3 (7.4)	34/12 50/19	17/29 30/39	Grade I and II Grade I and II	1 days-3 days
Dejan 2021 [44]	Conventional therapy Physical therapy	1. Age between 18 and 30 years 2. Weight within normal range, no major chronic or acute diseases 3. No history of acute ankle sprain in the past 6 months	RICE Hydrotherapy with hydrogen-rich water (HRW)	9 9	23.7 (4.0)	18/0	NA NA	Grade I and II Grade I and II	24 h 24 h
Muhammad 2015 [45]	Conventional therapy Physical therapy	1. Complete ligament rupture 2. Joint instability 3. Positive anterior drawer test 4. Under 40 years of age 5. Presenting within 48 h after injury	Plaster of Paris (POP) Functional therapy	60 60	29.83 (6.30) 28.77 (6.72)	18/42 25/35	39/21 30/30	Grade I and II Grade I and II	2w-6w 2w-6w
Chris 2010 [46]	Conventional therapy Physical therapy	1. Age between 16 and 65 years 2. Acute grade 1 or grade 2 ankle sprains (< 7 days)	PRICE Early therapeutic exercise	51 50	26.6 (7.5) 25.3 (8.3)	34/17 35/15	NA NA	Grade I and II Grade I and II	1w-16w 1w-16w
Punt 2016 [47]	Conventional therapy Physical therapy	1. Age between 18 and 64 years 2. With mild (grade I) or moderate (grade II) lateral ankle sprain 3. Radiographic confirmation of ankle sprain in the emergency department	No treatment Physical therapy	30 30	33.5 (9.5) 34.7 (11.3)	12/18 20/10	20/10 8/22	Grade I and II Grade I and II	6 weeks 6 weeks
Nyanzi 1999 [32]	Conventional therapy Physical therapy	1. Injury within 100 h 2. Able to follow instructions 3. Age between 14 and 65 years	Placebo therapy Ultrasound therapy	25 26	15-65 14-63	17/8 13/13	NA NA	NA NA	1 days-14 days 1 days-14 days
Neha 2021 [48]	Conventional therapy Physical therapy	1. Age between 18 and 60 years 2. Exclusion of grade 3 ankle sprains, lower limb fractures, chronic ankle injuries, open ankle wounds, contraindications to manual therapy or tape application	Placebo therapy Exercise therapy	16 16	28.4 (7.0) 26.1 (6.6)	10/6 9/7	8/8 8/8	Grade I and II Grade I and II	2w-1-6 m 2w-1-6 m

Table 1 (continued)

Research	Group	Inclusion Criteria	Intervention	Participants	Age	Sex (M/F)	Side (Left/Right)	Type	Follow up
James 2009 [49]	Conventional therapy	1. Acute varus ankle sprain occurring within 96 h prior to the initial physical therapy 2. Accompanied by swelling or tenderness	Conventional therapy	7	17.3 (5.1)	3/4	4/3	Grade I and II	2w-4w-6w
	Physical therapy		Stretching therapy	7	22.6 (14.3)	3/4	4/3	Grade I and II	2w-4w-6w

BMI, body mass index

(ICTRP) and ClinicalTrials.gov were searched for unpublished trials. Grey literature was identified through OpenGrey and ProQuest Dissertations & Theses. The search strategy combined controlled vocabulary and free text terms related to acute ankle sprain, physical therapy, and conventional treatment. The complete search strategies for all databases are provided in Appendix 1, including Boolean operators, truncations, and field tags, in line with PRISMA-S guidelines.

Backward citation tracking (reference lists of included studies) and forward citation tracking (using Scopus and Google Scholar) were performed to identify additional eligible studies.

Data extraction and quality control

Literature search and data extraction were independently performed by two investigators (Yang Changsen and Jia Zhengfeng). The full texts of potentially eligible articles were retrieved and assessed for eligibility against the PICOS criteria. Any discrepancies were resolved through discussion or validation by a third investigator.

Data extraction

A standardized data extraction form was used to collect study characteristics (author, year, country, sample size, population demographics, injury grade, intervention and comparator details, follow-up duration) and outcome data. Two reviewers independently extracted the data and cross-checked entries for accuracy.

Risk of bias assessment

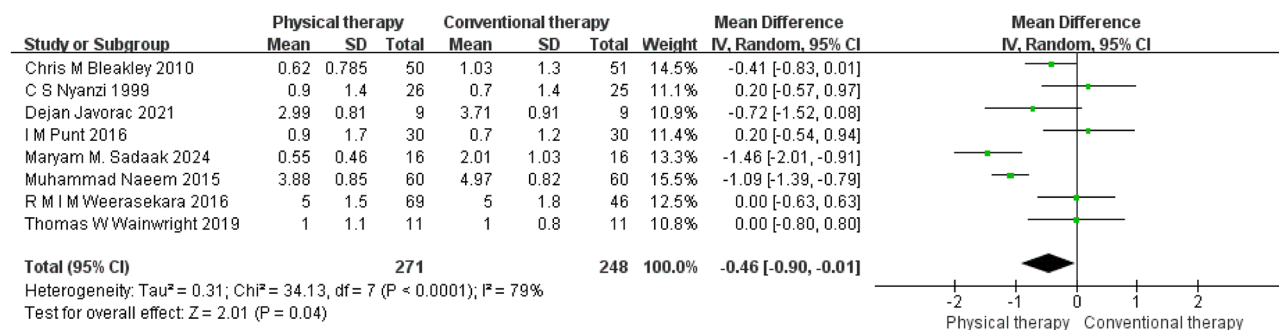
The risk of bias for each included RCT was independently assessed by two reviewers using the Review Manager tool across the following domains: random sequence generation, allocation concealment, blinding of participants and personnel, blinding of outcome assessment, incomplete outcome data, selective reporting, and other bias. Discrepancies were resolved by consensus.

Certainty of evidence

The certainty of the evidence for the study outcomes was evaluated using the Grading of Recommendations Assessment, Development and Evaluation (GRADE) approach. The results are summarized in a Summary of Findings (SoF) tables.

Data synthesis and statistical analysis

Data were organized using Microsoft Excel, and then meta-analysis of outcome data, sensitivity analysis, and publication bias assessment were performed using the Cochrane Collaboration’s Review Manager (version 5.4) and StataMP 17 statistical software. Trial sequential analysis (TSA) was performed using TSA 0.9.5.10 beta software. Statistical significance was set at $P<0.05$.

**Fig. 2** Post-treatment VAS scores

Continuous outcome variables were analyzed using the inverse variance (IV) method to calculate weighted mean differences (WMD) and 95% confidence intervals (CI). Dichotomous outcome variables were analyzed using the Mantel-Haenszel (M-H) method to calculate the odds ratios (OR). The choice between the fixed-effects model and the random-effects model was determined using the I^2 statistic and Q-test results. If I^2 was $< 50\%$ or the P -value of the Q-test was > 0.05 , indicating low heterogeneity among the included studies, a fixed-effects model was used for meta-analysis. If I^2 was $> 50\%$ or the P -value of the Q-test was < 0.05 , indicating high heterogeneity, a random-effects model was used. Sensitivity analyses were performed using a leave-one-out approach to assess any undue impact of any individual study on the pooled effect size. Publication bias was evaluated using Begg's and Egger's tests. TSA was conducted to assess the required information size and control the risk of type I and type II errors.

Results

Basic characteristics of included studies and patients

The literature search yielded 2891 articles, which were narrowed down to 1995 after removing duplicates. After title and abstract screening, 515 articles remained, and full-text review further reduced the number to 23. Following a detailed assessment for meta-analysis suitability, topic relevance, and exclusion of review articles, 10 articles were ultimately included. All included studies were conducted outside of China. A PRISMA 2020 flow diagram is presented in Fig. 1. The list of studies excluded

after full-text review, along with reasons for exclusion, is presented in Supplementary Table 1. The ten RCTs included in this study involved a total of 565 patients with grade I or II acute ankle sprains, of which 294 patients received physical therapy and 271 received conventional treatment. The duration of follow-up in the included studies varied from a few days to a few weeks.

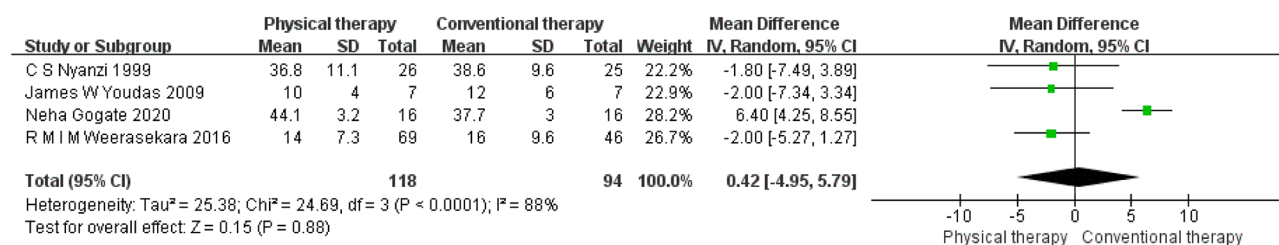
Physical therapy included exercise therapy, ultrasound therapy, thermotherapy, hydrotherapy, stretching, and functional therapy. Conventional treatments included RICE treatment, standard treatment, placebo, and cast immobilization (Table 1).

Risk of bias assessment

The VAS pain score was reported in eight studies and ROM activity in four studies, both of which were assessed across seven core domains of bias (including selection bias and implementation bias). The results of the risk of bias assessment are provided in Supplementary Figs. 1 and 2. Owing to the limited number of studies in the ROM group ($n = 4$), the statistical power of risk of bias assessment may be insufficient, and the findings should therefore be interpreted with caution.

Summary of outcome variables

The included studies assessed two primary outcomes: pain levels using the VAS and ankle mobility through dorsiflexion ROM. Subgroup analyses further examined VAS scores at rest and VAS scores during exercise after appropriate treatment.

**Fig. 3** Dorsiflexion range of motion (ROM)

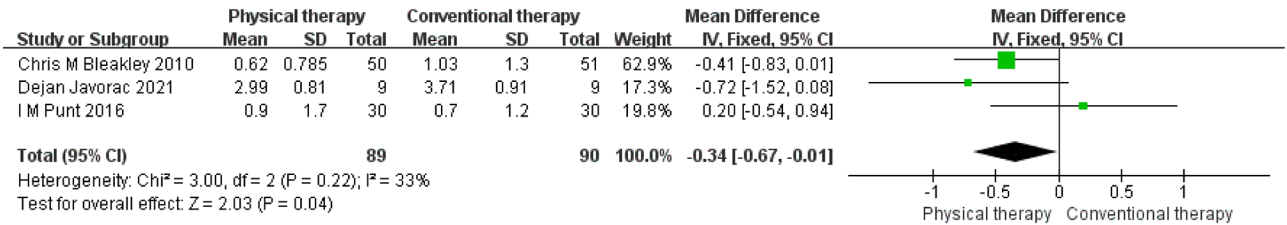


Fig. 4 VAS scores at rest

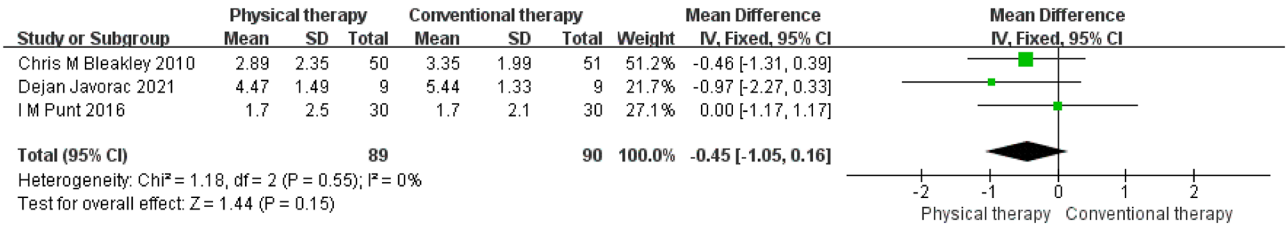


Fig. 5 VAS scores during exercise

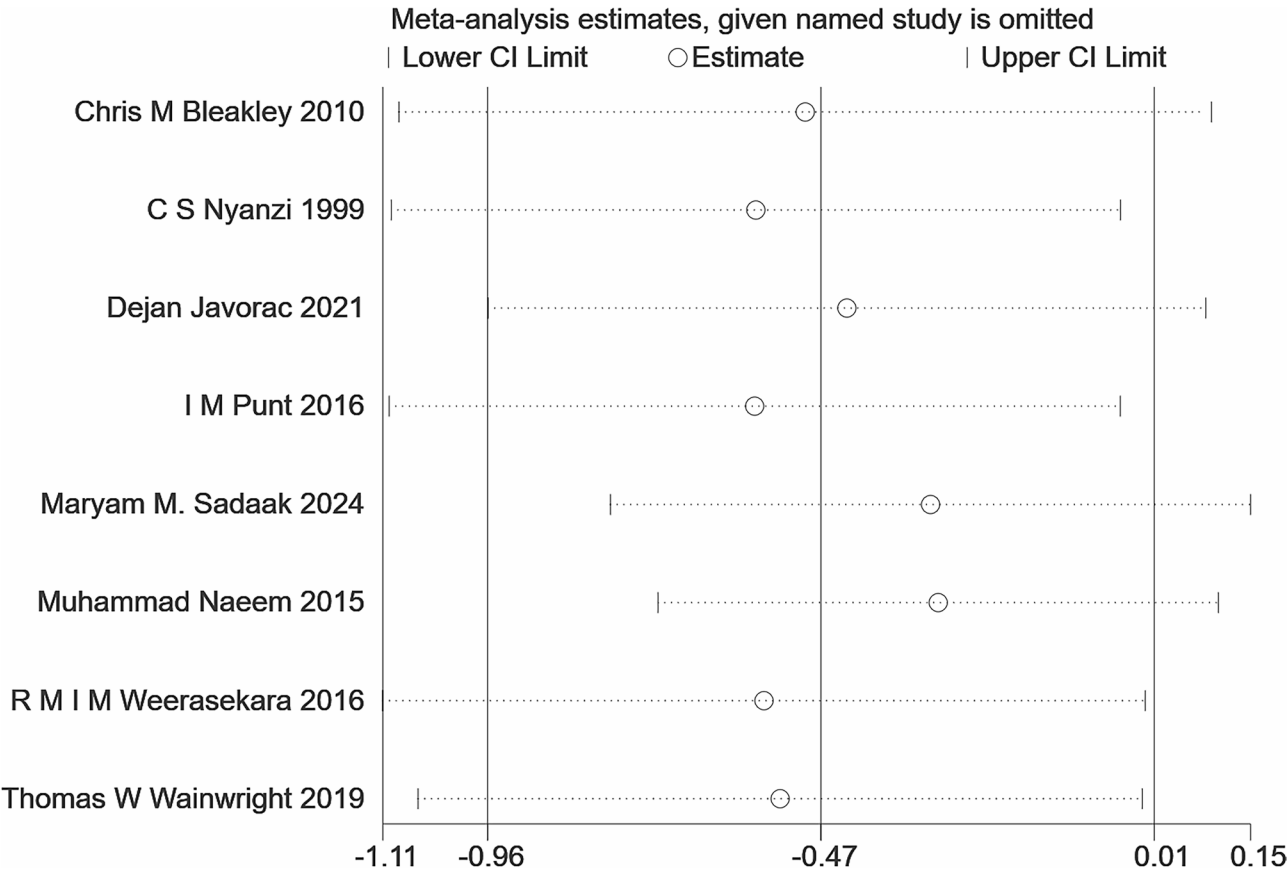


Fig. 6 Results of the sensitivity analysis

VAS scores
Eight of the included studies addressed the VAS scores of patients after treatment. Meta-analysis of these 8 studies revealed that patients with grade I and II acute ankle sprains who received physical therapy had significantly lower VAS scores compared to those receiving

conventional treatment (WMD = -0.46, 95% CI = -0.90 to -0.01, P = 0.04; I² = 79%; Fig. 2) (Certainty of Evidence: High; Supplementary Table 2).

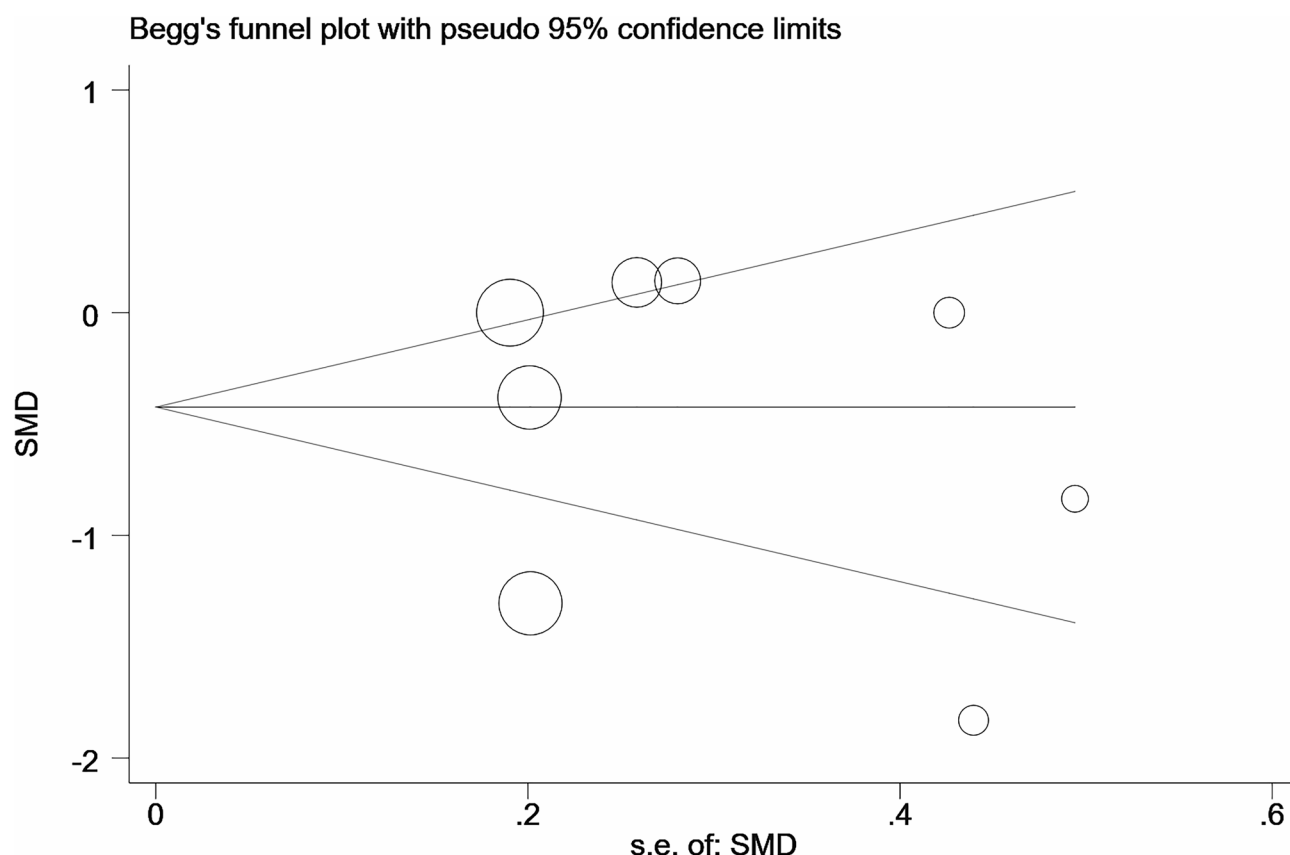


Fig. 7 Results of Begg's test

Dorsiflexion ROM

Four of the included studies had investigated post-treatment dorsiflexion ROM. Meta-analysis of these 4 studies revealed no significant difference in dorsiflexion ROM between patients with grade I or II acute ankle sprains who received physical therapy versus conventional treatment (WMD = 0.42, 95% CI = -4.95 to 5.79, $P = 0.88$; $I^2 = 88\%$; Fig. 3) (Certainty of evidence: Moderate; Supplementary Table 3).

Subgroup analysis

VAS score during rest

Three RCTs investigated post-treatment VAS scores at rest. Meta-analysis of these 3 studies revealed that patients with grade I and II acute ankle sprains who received physical therapy had significantly lower VAS scores at rest compared to those who received conventional treatment (WMD = -0.34, 95% CI = -0.67 to -0.01, $P = 0.04$; $I^2 = 33\%$; Fig. 4) (Certainty of evidence: High; Supplementary Table 4).

VAS scores during exercise

Three of the included RCTs investigated post-treatment VAS scores of patients during exercise. Meta-analysis of these 3 studies revealed no significant difference in VAS

scores during exercise between patients with grade I and II acute ankle sprains who received physical therapy versus conventional treatment (WMD = -0.45, 95% CI = -1.05 to 0.16, $P = 0.15$) ($I^2 = 0\%$; Fig. 5) (Certainty of evidence: High; Supplementary Table 5).

Sensitivity analysis

Sensitivity analysis of the VAS score forest plot showed that removing any one of the 8 included studies did not significantly alter the overall results, indicating that no single study had excessive influence on the pooled effect size (Fig. 6).

Assessment of publication bias

Publication bias analysis was conducted on the 8 studies examining post-treatment VAS scores. Both Begg's test ($z = 1.11$, $P > 0.05$) and Egger's test ($t = -0.32$, $P > 0.05$) indicated no significant publication bias, suggesting that the results of the meta-analysis were unlikely to have been affected by selective reporting (Figs. 7 and 8).

Trial sequential analysis

The results of the TSA indicated that the Z curve crossed the trial sequential monitoring boundary and the required information size (RIS), indicating that the

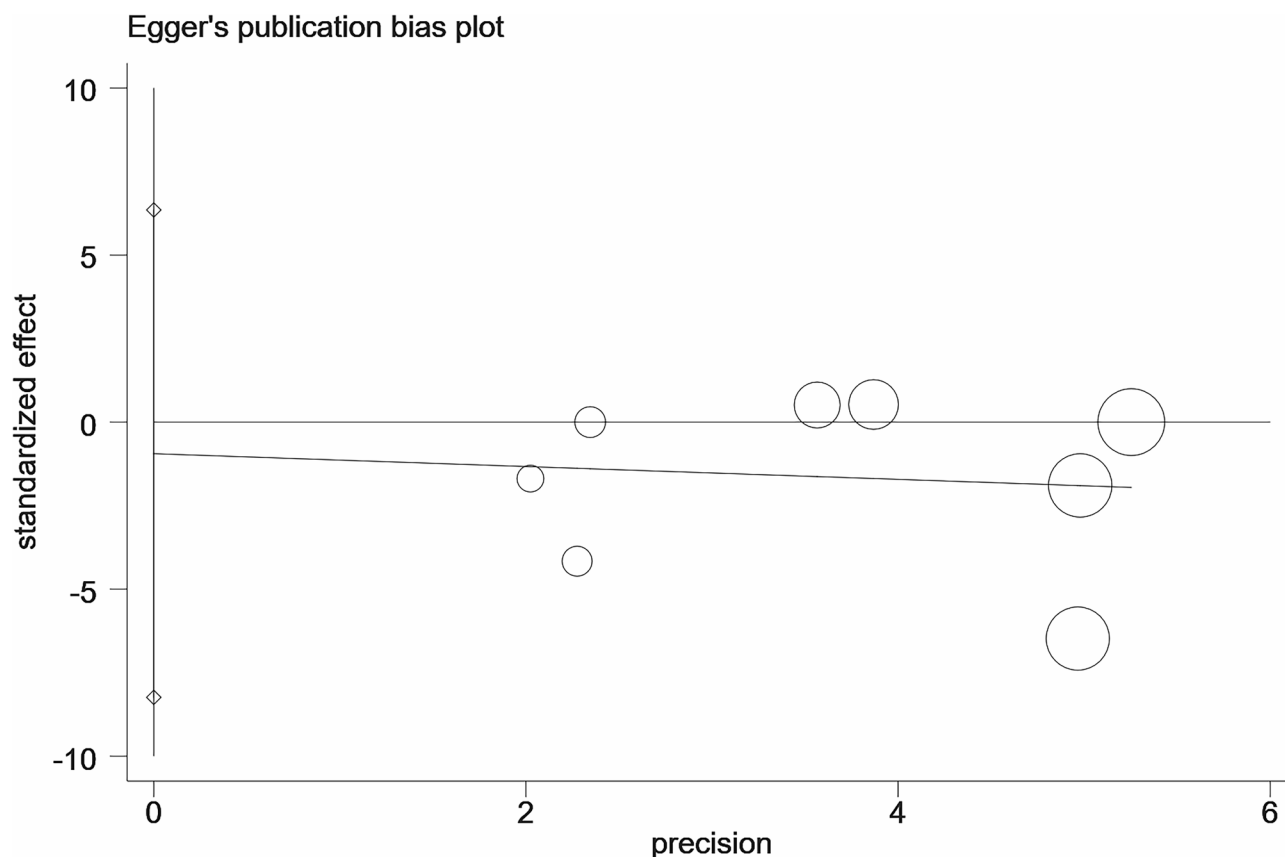


Fig. 8 Results of Egger's test

sample size was sufficient. Thus, it can be inferred that physical therapy significantly reduces VAS scores compared to conventional treatment (Fig. 9 and Supplementary Figs. 3–5).

Discussion

This meta-analysis of 10 RCTs found that physical therapy significantly outperformed conventional treatment in reducing pain, particularly at rest, for patients with grade I and II acute ankle sprains. However, no significant difference was observed between the two treatment groups with respect to improvement in ankle dorsiflexion mobility. The measurement methods for dorsiflexion ROM vary across studies, with some reporting active ROM (AROM) and others passive ROM (PROM), under either weight-bearing or non-weight-bearing conditions. Additionally, specific measurement techniques—such as goniometers, inclinometers, or visual estimation—are reported inconsistently, contributing to methodological heterogeneity. The findings suggest that physical therapy is an effective treatment option, especially for patients who need to quickly regain their ability to perform activities of daily living and mobility. However, conventional treatments (e.g., RICE therapy and cast immobilization)

may still be suitable for patients requiring greater joint mobility.

The significant benefits of physical therapy in pain relief for acute ankle sprains may be attributed to its multimodal approach, which combines methods like exercise therapy, ultrasound therapy, temperature therapy, hydrotherapy, stretching, and functional therapy [26–28]. This comprehensive approach promotes blood circulation, reduces local inflammatory responses, improves muscle strength, and enhances joint stability, ultimately leading to effective pain reduction [29–31]. Previous studies, such as those by Weerasekara et al. and Nyanzi et al. [31, 32] support the effectiveness of ultrasound and temperature therapy in reducing pain after ankle sprains, which is consistent with our results. Additionally, studies by Bleakley et al. and Mattacola et al. highlight the benefits of exercise therapy and functional training in enhancing muscle strength and joint stability, contributing to pain reduction [33, 34].

Physical therapy's benefits may also extend to psychological aspects, as the support and social interaction inherent in therapy can positively impact pain perception. Besides being a physical injury, an acute ankle sprain may also induce psychological stress and anxiety. Studies suggest that psychological support can significantly

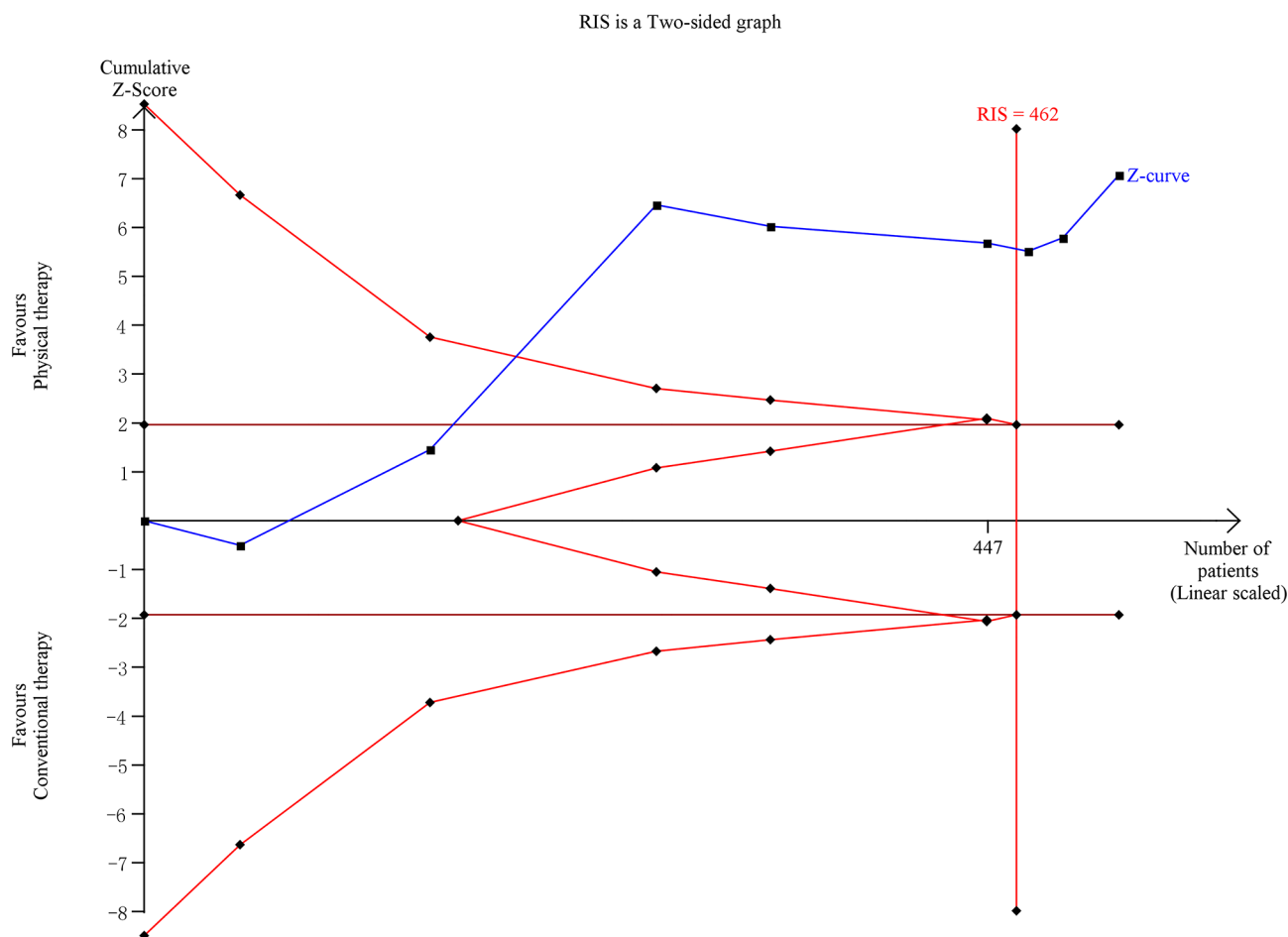


Fig. 9 Results of Trial Sequential Analysis

improve patients' pain perception [35, 36]. Therefore, physical therapy's comprehensive approach may benefit patients on both physiological and psychological levels. In addition, patient education—considered the cornerstone of physical therapy—plays a key role in promoting treatment compliance, managing patient expectations, and alleviating injury-related anxiety. Education regarding proper exercise techniques, activity modifications, and self-management strategies can further enhance the functional outcomes of physical therapy.

Physical therapy's limited impact on improving ankle dorsiflexion mobility may be due to lingering joint swelling, which takes longer to subside despite pain relief [37]. In addition, certain physical therapies (e.g., ultrasound therapy and hydrotherapy) focus more on pain relief rather than joint mobility enhancement [38]. Studies by Weerasekara et al. and Tennakoon et al. [31] support these findings, suggesting that physical therapy's benefits may be more pronounced for pain management rather than joint mobility improvement.

Subgroup analysis showed physical therapy was more effective in reducing VAS scores at rest, validating its

pain management benefits in daily life. However, VAS scores during exercise did not differ significantly between the two groups, possibly due to exercise-induced pain perception, which may persist even after physiotherapy [39].

This study's use of sensitivity analysis and TSA provided robust evidence for physical therapy's benefits in pain relief for grade I and II acute ankle sprains. The findings align with previous studies, such as those by Zhao et al. [40] and Bruno et al. [41], demonstrating physical therapy's effectiveness in reducing pain. Additionally, the results support the study by Brison et al. [42], which found no significant difference in joint mobility between physical therapy and conventional treatment.

Some limitations of this study should be acknowledged. These include methodological heterogeneity due to variations in physical therapy methods and conventional treatment protocols across studies, inconsistent follow-up times, and reliance on subjective pain assessment tools like the VAS. Future research should aim to standardize treatment protocols and follow-up times, and consider incorporating more objective assessment measures, such

as ultrasound imaging and joint mobility measurements, to reduce bias.

The heterogeneity of dorsiflexion measurement methods across RCTs represents a significant limitation. Variability in the choice between active or passive ROM, weight-bearing versus non-weight-bearing conditions, and the use of different measurement tools may contribute to inconclusive results regarding changes in dorsiflexion. This methodological inconsistency complicates the interpretation of physical therapy's effectiveness in improving joint ROM and highlights the need for standardized measurement protocols in future studies. Furthermore, the use of immobilization itself may have restricted mobility recovery, potentially explaining the lack of significant difference in mobility between the two treatments. However, this does not negate the advantages of physical therapy in terms of pain relief.

The variability in the timing of VAS measurements across the included studies (ranging from 1 week to 6 weeks post-treatment) introduces heterogeneity, which may limit the generalizability of the results. Due to differences in healing rates, treatment frequencies, and patient compliance, pain levels may vary significantly within this time frame. This variability complicates direct comparisons and underscores the need for standardized follow-up durations in future RCTs.

Conclusion

Physical therapy significantly outperforms conventional treatment in reducing VAS pain scores in patients with grade I and II acute ankle sprains, particularly at rest. However, it did not show a significant advantage in improving ankle dorsiflexion mobility. Physical therapy can be an effective treatment option, especially when rapid pain relief is required. Future studies should focus on optimizing treatments for mobility improvement and include larger sample sizes with longer follow-up periods for more robust evidence.

Abbreviations

RCTs	Randomized controlled trials
VAS	Visual analogue scale
ROM	Range of movement
TSA	Trial sequential analysis
RICE	Rest, ice, compression, elevation
WMD	Weighted mean difference

Supplementary Information

The online version contains supplementary material available at <https://doi.org/10.1186/s13018-025-06272-3>.

Supplementary Material 1
Supplementary Material 2
Supplementary Material 3
Supplementary Material 4
Supplementary Material 5

Supplementary Material 6

Supplementary Material 7

Supplementary Material 8

Supplementary Material 9

Supplementary Material 10

Supplementary Material 11

Acknowledgements

Not applicable.

Author contributions

JL: Conceptualization; ZJ, WG, and CY: Study design, data collection, and analysis; ZJ and CY: Original manuscript writing; WG and ZJ: Language check, review & editing. All authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

Funding

This work was supported by National Natural Science Foundation of China (Grant 82302788) and the Special Funding of the National Clinical Research Center for Orthopedics, Sports Medicine and Rehabilitation.

Data availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate

This study is a review and meta-analysis with no ethical implications.

Consent for publication

Not applicable.

Competing interests

The authors declare no competing interests.

Received: 7 May 2025 / Accepted: 29 August 2025

Published online: 29 September 2025

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